Health and safety education

How can education contribute to a healthier and safer society?

Dr Wonyong Park | EDUC1033 Week 10 · Lecture



School crossing patrol · Pauline E, geograph.org.uk, CC BY-SA 2.0

Session outline

1 Health, wellbeing and health literacy

2 Health education in England and Southampton LifeLab

3 Risk and school health and safety

4 Comprehensive school safety and climate disruption

Activity 1

In pairs | 3 minutes

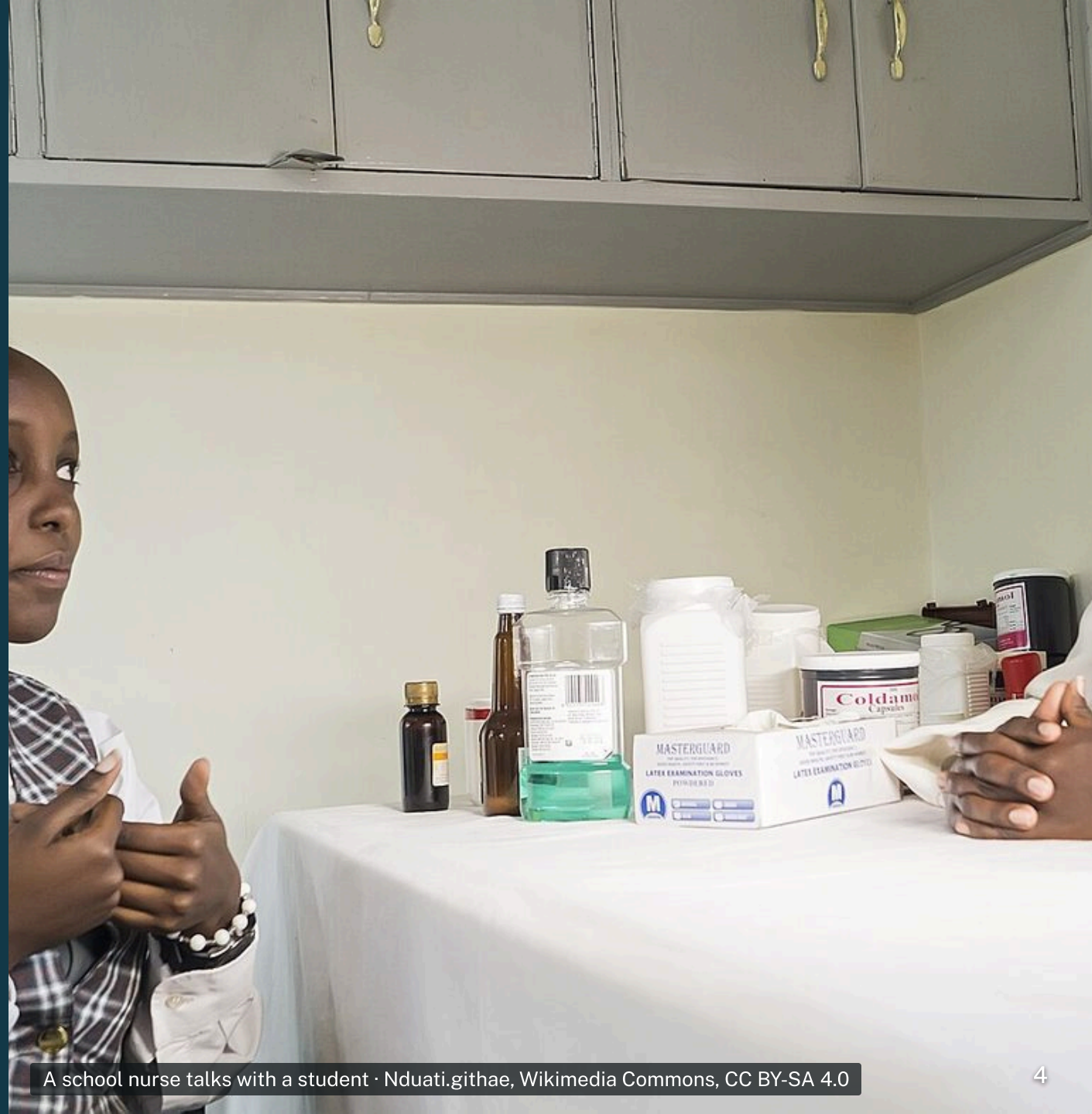
Think of one thing you were taught at school about staying healthy or safe.

Did it change what you did? If not, what got in the way?

Keep your example: we come back to it at the end.

Health, wellbeing and health literacy

What counts as health, why health promotion is wider than information, and why knowing is not the same as doing.



A school nurse talks with a student · Nduati.githae, Wikimedia Commons, CC BY-SA 4.0

A broad definition of health

“Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.”

World Health Organization (WHO, 1946), Constitution, preamble

What it adds

- Body, mind and relationships all belong to health
- Loneliness, exclusion and insecurity become health questions
- Health as a right: the constitution names the highest attainable standard of health as a fundamental human right

Where it is hard to apply

- "Complete" wellbeing would make anyone with a disability or long-term condition unhealthy
- A pupil with a well-managed condition may be thriving
- Wellbeing is not constant happiness: guidance in England separates ordinary worry or sadness from mental health conditions

WHO (1946), Constitution; Department for Education (DfE, 2025), pp. 20–21, 27.

Health promotion is wider than giving information

“Health promotion is the process of enabling people to increase control over, and to improve, their health.”

WHO (1986), Ottawa Charter for Health Promotion

- Build healthy public policy
- Create supportive environments
- Strengthen community action
- **Develop personal skills**, where health education mainly sits
- Reorient health services

A school can teach about nutrition **and** look at the food it serves; it can teach emotional awareness **and** make trusted support easy to reach.

A young person may want to be more active but lack affordable options or feel unsafe outdoors. Repeating the advice adds little unless those barriers are addressed.

WHO (1986), Ottawa Charter for Health Promotion.

Three forms of health literacy

Nutbeam (2000, 2025) distinguishes three forms. Each points to a different aim for health education.

Functional

Understanding and using basic health information.

For example: reading a food label, recognising a warning, following a clear instruction.

Interactive

Asking questions, discussing options and seeking support.

For example: explaining a worry to a trusted adult, or asking a pharmacist about a medicine.

Critical

Evaluating claims and the conditions that shape health.

For example: comparing an influencer's claim with research evidence, or spotting a commercial interest.

Nutbeam (2000), *Health Promotion International* 15(3); Nutbeam (2025), *Health Promotion International* 40(4).

Why knowing may not change what people do: COM-B

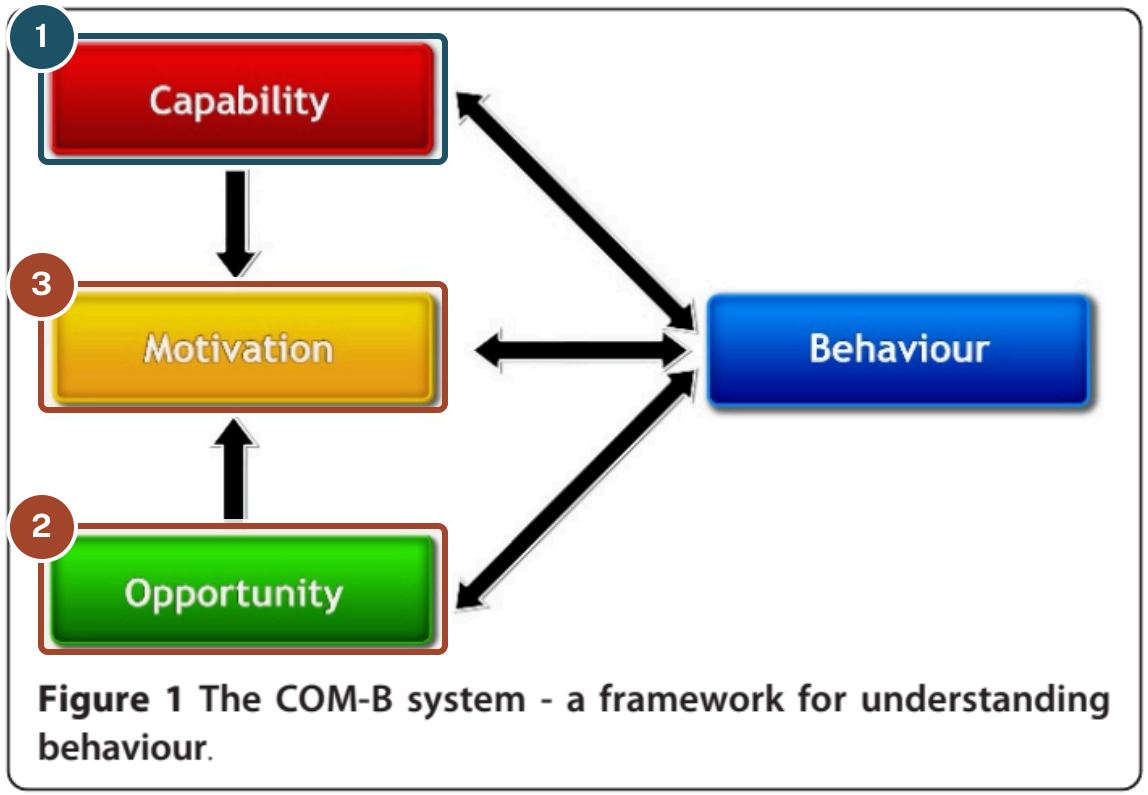


Figure 1 The COM-B system - a framework for understanding behaviour.

Behaviour depends on capability, opportunity and motivation, which also affect each other. Example: eating a nutritious meal.

1. **Capability:** knowing how to prepare the meal. This is where lessons mostly act.
2. **Opportunity:** access to ingredients, time and cooking facilities.
3. **Motivation:** enjoyment, expectations and habit.

Michie, van Stralen & West (2011), Figure 1, p. 4. CC BY 2.0.

The health-promoting school: making lessons usable

WHO and UNESCO (2021) standards connect

- **Curriculum:** what pupils learn about health
- **Environment:** food, activity, relationships, routes to support
- **Governance:** policies and staff capacity
- **Partnerships:** families and local services

A Cochrane review of the approach

- **67** cluster-randomised trials
- **Small improvements** in some outcomes: physical activity, fruit and vegetable intake, tobacco use, reports of being bullied
- Evidence quality **low to moderate**
- No clear benefit for every health outcome, or for academic achievement

WHO & UNESCO (2021), *Making every school a health-promoting school*; Langford et al. (2014), Cochrane Review CD008958.

Health education in England and Southampton LifeLab

What schools in England must teach from September 2026, and what one local programme and its trials show.



Southampton General Hospital · David Martin, geograph.org.uk, CC BY-SA 2.0

RSHE: the statutory framework in England

July 2025 — The Department for Education (DfE) publishes revised statutory guidance on **relationships, sex and health education (RSHE)**.

1 Sept 2026 — The revised guidance **comes into effect** in schools.

Scope — **England only.** Scotland, Wales and Northern Ireland have their own curriculum arrangements.

What is compulsory, and where

- **Relationships education:** primary
- **Relationships and sex education (RSE):** secondary
- **Health education:** state-funded schools, including academies and free schools
- **Independent schools:** a separate requirement for personal, social, health and economic (PSHE) education

DfE (2025), *Relationships Education, RSE and Health Education*, introduction; DfE (2026), publication page.

What health education covers

Primary foundations

- Emotional vocabulary and mental wellbeing
- Physical activity, nutrition, sleep and hygiene
- Personal safety: fire, roads, railways and water
- Basic first aid

Secondary progression

- Evaluating online information; using health services
- Peer influence, exploitation, conflict and violence
- Health protection, substances, developing bodies
- First aid including CPR and the purpose of defibrillators

The guidance "aims to support and not to alarm pupils" (DfE, 2025, p. 5).

DfE (2025), pp. 5, 20–31.

Southampton LifeLab: practical science about health

LifeLab links schools with the University of Southampton and a hospital-based learning environment. Pupils use specialist equipment and meet researchers working on current health questions.

DNA extraction

Making an abstract idea visible and hands-on.

Ultrasound of blood flow

Measuring blood flow through the carotid artery.

Development before birth

How early life connects with health later on.

The stated aim: confidence, self-efficacy and pupils' own reasons for action, rather than following instructions.

University of Southampton (n.d.), LifeLab; Grace (n.d.), slides 22–31.

LifeLab: the programme director's overview



https://www.youtube.com/watch?v=aBLZLUzK2BU · University of Southampton, *LifeLab, University of Southampton* (4 min)

The 2021 trial: a lasting gain in health literacy

Cluster-randomised trial in **38 schools**: 2,929 adolescents at the start, 2,487 followed up after 12 months.

Table 2. Change in the primary outcome of theoretical health literacy score.

Primary Outcome		β (adjusted difference*)	95%CI
Standardised Total Theoretical Health Literacy	SD units	0.27	(0.12, 0.42)

1. LifeLab schools scored **0.27 standard deviations** higher on theoretical health literacy, after adjustment.

2. The 95% confidence interval runs from **0.12 to 0.42**, so it does not include zero.

3. Reported **diet quality** did not improve.

Woods-Townsend et al. (2021), *PLOS ONE*, Table 2 (excerpt: primary outcome row), p. 10. CC BY 4.0.

EACH-B: adding teacher support and an app

Engaging Adolescents in Changing Behaviour (EACH-B) combined three components.

- A LifeLab visit
- Healthy Conversation Skills for teachers
- A gamified smartphone app

Outcome trial (Barker et al., 2026)

- **45 schools**; 2,065 students at the start
- **55%** followed up after one year
- No demonstrated improvement in the main **diet** or **physical activity** outcomes

Process evaluation (Jenner et al., 2025)

- Students and teachers **valued LifeLab**
- Engagement with the **app was limited**
- Teachers met practical barriers to health conversations

Barker et al. (2026), NIHR programme report; Jenner et al. (2025), *BMJ Public Health* 3(2).

What the LifeLab evidence supports

Shown in a trial

Educational outcome

Theoretical health literacy improved and lasted twelve months.

Not shown

Behavioural outcome

Diet and physical activity did not improve in either trial.

Not measured

Health outcome

Effects on later disease would need long follow-up.

Understanding evidence, confidence in asking questions and critical reflection are educational gains in their own right.

Woods-Townsend et al. (2021); Barker et al. (2026); Jenner et al. (2025).

Risk and school health and safety

Hazard and risk, who is responsible in schools, and how to control risk without removing every challenge.



Fire drill, Baozhong Junior High School, Taiwan · Wikimedia Commons, CC BY 2.0

Hazard, risk and safety

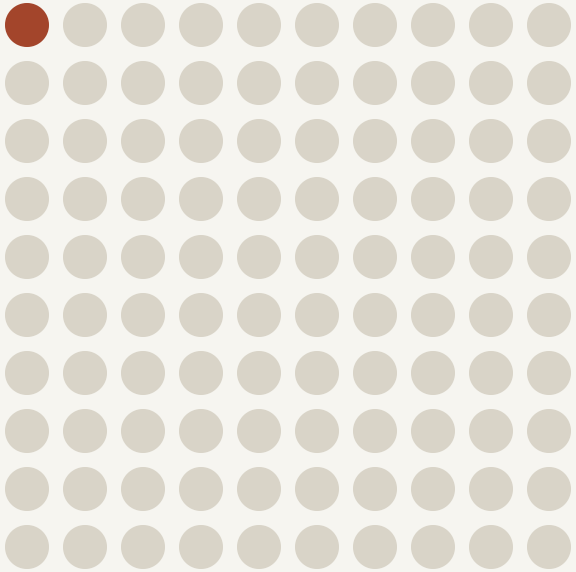
Hazard	A potential source of harm, such as water.
Risk	The chance and consequences of harm in particular circumstances: weather, supervision, swimming ability, help nearby.
Disaster risk	The interaction of a hazard with **exposure, vulnerability and capacity**. Social conditions are part of the explanation.
Safety	Significant risks are managed. It does not mean harm is impossible.

Feeling safe and being safe can differ: vivid news can make rare dangers seem common, and familiar routines can seem safer than they are.

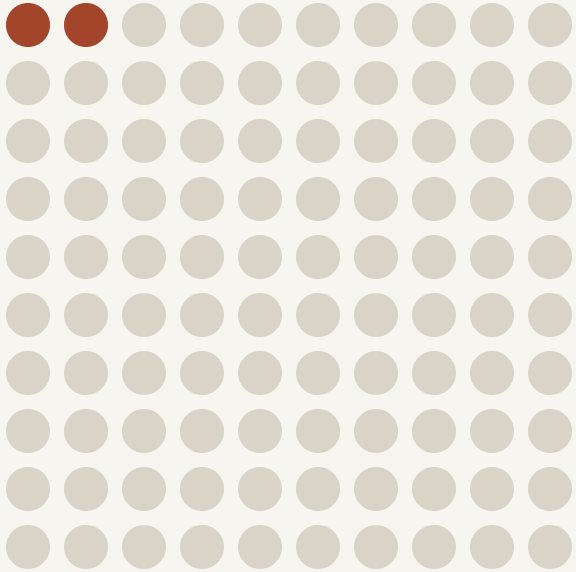
United Nations Office for Disaster Risk Reduction (UNDRR, 2017), terminology: disaster risk.

Reading a risk claim: relative and absolute change

Before: 1 in 100



After: 2 in 100



+100%	**Relative change:** "risk doubles"
+1 point	**Absolute change:** one more case in every hundred

Ask of any risk claim: **which population**, **over what period**, and **which outcome**?

Hypothetical example.

Who is responsible for health and safety in schools?

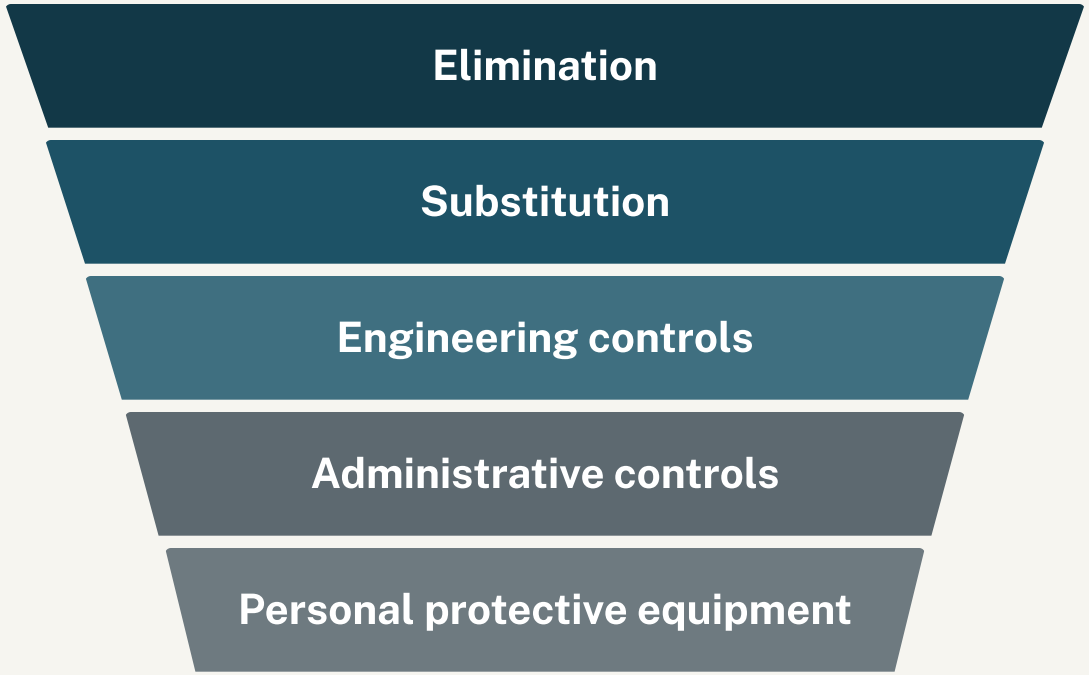
Employer or proprietor	Accountable for health and safety. In independent schools, this is the proprietor.
School leaders	Usually run day-to-day arrangements, with competent advice, policies and staff training.
Risk assessment	Identify hazards, who might be harmed and suitable controls; record significant findings simply; review when things change.
After incidents	Investigate, share lessons and improve arrangements.

Pupils can help spot problems, but knowing a rule cannot make up for unsafe conditions.

DfE (2022), *Health and safety: responsibilities and duties for schools*, sections 1–5, 12–13.

Controlling risk: change the setting before relying on behaviour

The hierarchy of controls



Most reliable at the top

DfE guidance for public health incidents sets out these levels and warns against simply choosing the **easiest** measure.

A reminder to be careful depends on everyone, every time. Reducing exposure through the environment protects everyone.

A school crossing: pupils learn to judge traffic, and their observations of blocked sightlines can go to the authorities who can change the road.

DfE (2022), section 5.

Weighing risks and benefits

“the goal is not to eliminate risk, but to weigh up the risks and benefits”

Health and Safety Executive (HSE, 2012), joint statement with the Play Safety Forum

Beneficial challenge

- Climbing, exploring, school visits
- Build skills, judgement and independence
- Plan in proportion: transport, supervision, access, specialist advice

Preventable hazard

- Poorly maintained equipment
- Serious risks with no learning benefit
- Should be fixed, not accepted

Adjust activities so pupils with different abilities can take part, rather than excluding them by default.

HSE (2012), *Children's play and leisure: promoting a balanced approach*; DfE (2022).

Comprehensive school safety and climate disruption

A global framework for school safety, climate-related disruption in 2024, and preparedness. This builds on Week 6.



Flooding during torrential rain, Baozhong Junior High School, Taiwan · Wikimedia Commons, CC BY 2.0

The Comprehensive School Safety Framework 2022–2030



Aims at "safe, equitable, and continuous access to a quality education for all" across every kind of hazard (GADRRRES, 2022, p. 2).

1. **Safer learning facilities:** sites, buildings, water and sanitation, ventilation, evacuation routes.
2. **School safety and educational continuity management:** risk assessment, preparedness, response, recovery.
3. **Risk reduction and resilience education:** for children, staff and communities.
4. **Foundation:** enabling systems and policies, with responsibilities, coordination and resources.

Global Alliance for Disaster Risk Reduction and Resilience in the Education Sector (GADRRRES, 2022), summary document, pp. 1–3.

Educational continuity means more than reopening

Can every pupil actually learn?



Assumptions about internet access, quiet space, language or adult help can exclude families.



The framework names barriers linked to disability, digital inequality and social exclusion.



Plans need accessible materials, several communication channels and contact with those the default misses.

Resilience and institutional change



Capacities to **anticipate, absorb, adapt and transform**.



Recurring overheating may need changes to buildings and routines, not only advice to pupils.

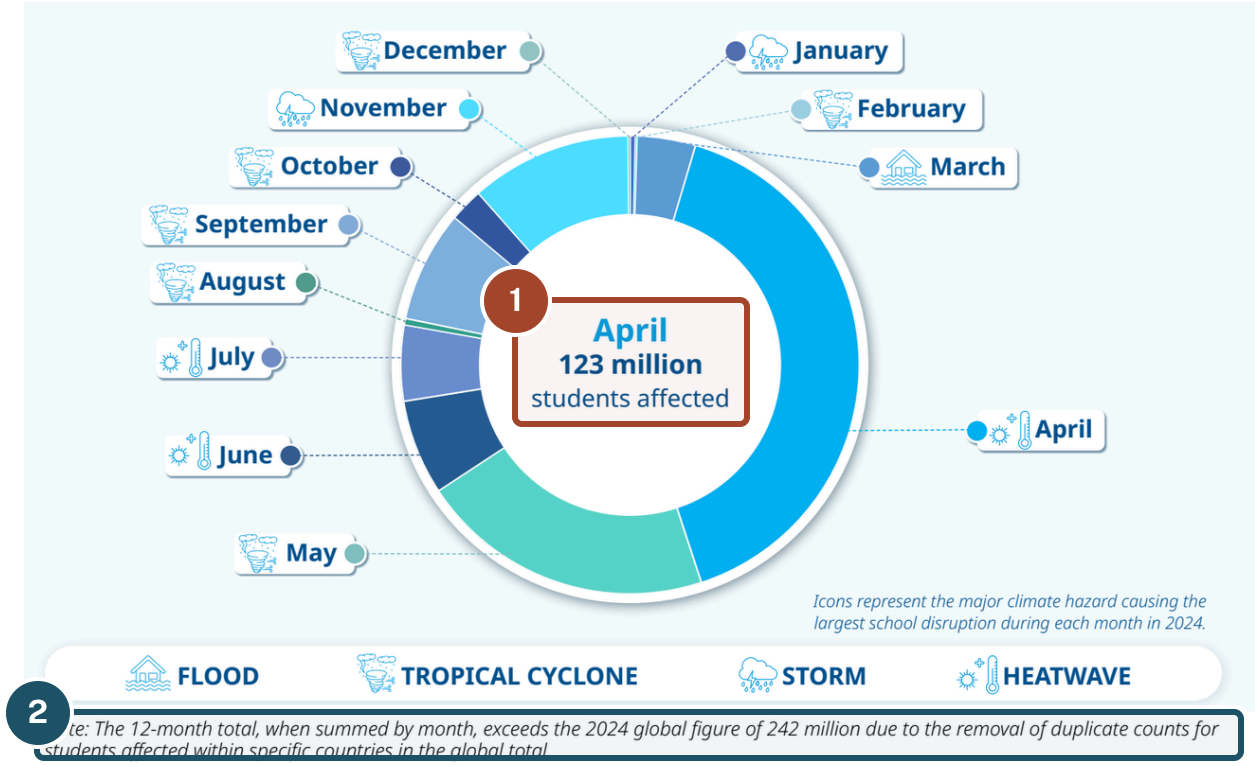


Resilience should not mean pupils and teachers putting up with unsafe conditions.

GADRRRES (2022), summary document.

Climate-related disruption to schooling in 2024

Figure 1. Number of students affected by climate-related school disruptions in 2024, by month.



Students affected by climate-related school disruption, by month. Icons show the main hazard each month.

1. **April** was the largest month: **123 million** students, mostly affected by heatwaves.

2. Monthly totals **cannot be added up**: some students were disrupted more than once. Without duplicates: at least **242 million** students in **85 countries**.

UNICEF (2025), *Learning interrupted: global snapshot of climate-related school disruptions in 2024*, Figure 1, p. 3.

Preparing children for climate-related disasters

How disasters harm children

- **Directly:** injury, illness, psychological effects
- **Indirectly:** disrupted care, separation, insecurity, lost education
- **Unequally:** age, disability, housing, income and support shape exposure and options

“isolated school-based programmes increase disaster knowledge, but behavioural change is not forthcoming”

Seddighi et al. (2020), p. 3. The same gap we saw with LifeLab.

What helps

- Drills that test the school's own arrangements, with debriefing
- Plans that work across languages, senses and mobility
- Confidence without alarm: a clear, achievable response

Seddighi, Yousefzadeh, López López & Sajjadi (2020), *BMJ Paediatrics Open* 4.

Different sources support different claims

TYPE OF SOURCE	WHAT IT CAN TELL US	THIS WEEK
Statutory guidance	What schools are expected to do	DfE (2025) RSHE; DfE (2022)
Framework	How to organise action	GADRRRES (2022); WHO & UNESCO (2021)
Review	What a body of studies suggests	Langford et al. (2014); Seddighi et al. (2020)
Trial	An intervention's effect under particular conditions	Woods-Townsend et al. (2021); Barker et al. (2026)
Process evaluation	What was delivered and how it was experienced	Jenner et al. (2025)

A broad framework is not proof that each part works; an inconclusive behaviour trial does not show that nothing was learned.

How can education contribute?

Capabilities

Health and risk literacy, practical judgement, and knowing how and where to seek help.

Conditions

Food, facilities, safe routes, risk controls and continuity plans that make advice usable.

Shared responsibility

Families, services and authorities act on what pupils notice. Children are not responsible for fixing public failures.

Education contributes most when informed judgement is backed by usable opportunities, institutional care and shared responsibility.

Back to your example

Return to your example from Activity 1.

1. Which part of COM-B did it target: capability, opportunity or motivation?
2. What in your school or community made it easier or harder to act on?
3. What evidence would show that it worked?

Further reading

- Woods-Townsend et al. (2021), *PLOS ONE*, LifeLab trial
- Nutbeam (2025), health literacy 25 years on
- GADRRRES (2022), Comprehensive School Safety Framework, summary
- DfE (2025), RSHE statutory guidance, pp. 20–31